

```
#include <stdio.h>

#define MAX 5

int stack[MAX];

int top = -1;

void push(int value) {
    if (top == MAX - 1) {
        printf("Stack Overflow! Cannot push %d\n", value);
    } else {
        top++;
        stack[top] = value;
        printf("%d pushed into stack.\n", value);
    }
}

void pop() {
    if (top == -1) {
        printf("Stack Underflow! Cannot pop.\n");
    } else {
        printf("%d popped from stack.\n", stack[top]);
        top--;
    }
}

void peek() {
    if (top == -1) {
        printf("Stack is empty. Nothing to peek.\n");
    } else {
        printf("Top element is: %d\n", stack[top]);
    }
}

void display() {
```

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if (top == -1) {
    printf("Stack is empty.\n");
} else {
    printf("Stack elements: ");
    for (int i = top; i >= 0; i--) {
        printf("%d ", stack[i]);
    }
    printf("\n");
}
}

int main() {
    int choice, value;
    while (1) {
        printf("\n--- Stack Menu ---\n");
        printf("1. Push\n2. Pop\n3. Peek\n4. Display\n5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter value to push: ");
                scanf("%d", &value);
                push(value);
                break;
            case 2:
                pop();
                break;
            case 3:
                peek();
                break;

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        case 4:
            display();
            break;
        case 5:
            printf("Exiting program.\n");
            return 0;
        default:
            printf("Invalid choice! Try again.\n");
    }
}
}

```

Output

```

--- Stack Menu ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:
1
Enter value to push: 10
10 pushed into stack.

--- Stack Menu ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter value to push: 20
20 pushed into stack.

--- Stack Menu ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter value to push: 30
30 pushed into stack.

```

```
Enter value to push: 30
30 pushed into stack.
```

```
--- Stack Menu ---
```

1. Push
2. Pop
3. Peek
4. Display
5. Exit

```
Enter your choice: 2
30 popped from stack.
```

```
--- Stack Menu ---
```

1. Push
2. Pop
3. Peek
4. Display
5. Exit

```
Enter your choice: 3
Top element is: 20
```

```
--- Stack Menu ---
```

1. Push
2. Pop
3. Peek
4. Display
5. Exit

```
Enter your choice: 4
Stack elements: 20 10
```

```
--- Stack Menu ---
```

1. Push
2. Pop
3. Peek
4. Display
5. Exit

```
Enter your choice: 5
Exiting program.
```

```
Process returned 0 (0x0)   execution time : 25.806 s
Press any key to continue.
```